

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



Mini Project Report

on

Smartguard Snatch Theft Detection And Auto-Alert System

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the Interdisciplinary Project Based Learning
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “**Smartguard Snatch TheftDetection And Auto-Alert System**” has been successfully carried out by **Vismaya P(1CR25AI142), Yash Krishna(1CR25AI143), Rakshita(1CR25CS139), Rakshith R K Reddy(1CR25CS140)** bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the, Interdisciplinary Project Based Learning Department of Basic Sciences.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide

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Abstract

The **SmartGuard Snatch Theft Detection System** is an intelligent security solution designed to detect and prevent snatch theft incidents in public and crowded areas. The system uses advanced technologies such as image processing, motion analysis, and real-time monitoring to identify suspicious activities related to bag, phone, or chain snatching. Cameras continuously capture video footage, and the system analyzes sudden movements, unusual speed, and victim-attacker interactions using machine learning algorithms. When a theft attempt is detected, SmartGuard immediately triggers alerts through alarms, mobile notifications, or security control systems, enabling quick response from authorities or nearby users. The system can also store captured images and video evidence for further investigation. By integrating smart surveillance with automated detection, the project improves public safety and reduces dependence on manual monitoring. SmartGuard is especially useful in places such as railway stations, bus stops, markets, shopping malls, and streets where snatch thefts are common. The proposed system offers high accuracy, faster response time, and enhanced security while remaining cost-effective and user-friendly. This project demonstrates how artificial intelligence and smart monitoring can contribute to safer urban environments.

Keywords:

- Snatch Theft Detection
- Smart Surveillance
- Image Processing
- Machine Learning
- Real-Time Monitoring
- Motion Detection
- Artificial Intelligence
- Security System
- Video Analytics
- Object Tracking
- Crime Prevention
- Automated Alert System
- CCTV Surveillance
- Suspicious Activity Detection
- Public Safety